



# SJAA EPHEMERIS

## SJAA Activities Calendar

Jim Van Nuland

(late September)

- 22 **General meeting at Houge Park.** 8 p.m. Slide and Equipment night.
- 29 **AANC-Con in San Mateo.**

### October

- 5 Astronomy Class at Houge Park. "Weather for Astronomers – tools to tell you whether to go", 7:30 pm.
- 5 Houge Park star party. Sunset 6:45 p.m., 21% moon rises 2:44 a.m. Star party hours: 8:00 to 11:00 p.m.
- 6 Dark sky weekend. Sunset 6:44 p.m., 13% moon rises 3:47 a.m.

### 11/13 CalStar star party at Lake San Antonio Park. See Page 5.

- 13 Dark sky weekend. Sunset 6:34 p.m., 8% moon sets 7:35 p.m.
- 19 Houge Park star party. Sunset 6:26 p.m., 59% moon sets 0:52 a.m. Star party hours: 7:30 to 10:30 p.m.
- 27 **General meeting at Houge Park.** 8 p.m. Peter Dunckel on amateur/professional cooperative research. Board of directors meets at 6 p.m.

### November

- 2 Astronomy Class at Houge Park. "Meet the Neighbors – a 3D view of the Milky Way", 7:30 p.m.
- 2 Houge Park star party. Sunset 6:09 p.m., 35% moon rises 1:40 a.m. Star party hours: 7:00 to 10:00 p.m.
- 3 Dark sky weekend. Sunset 6:08 p.m., 26% moon rises 2:42 a.m.
- 4 DST ends. Retard clock at 2 a.m. -> 1 a.m.
- 10 Dark sky weekend. Sunset 5:02 p.m., 2% moon sets 2:42 a.m.
- 16 Houge Park star party. Sunset 4:57 p.m., 43% moon sets 10:47 p.m. Star party hours: 7:00 to 10:00 p.m.
- 17 **General meeting at Houge Park.** 8 p.m.
- 18 **Fall Swap at Houge Park.** Noon until late afternoon
- 30 Astronomy Class at Houge Park. "The Local Group – A 3D view of what is around the Milky Way", 7:30 p.m.
- 30 Houge Park star party. Sunset 4:51 p.m., 53% moon rise 2:01 a.m. Star party hours: 7:00 to 10 p.m.

**24 hour news and information hotline:**

**(408) 559-1221**

### October General Meeting

**Peter Dunckel to speak at 8 p.m.**

**October 27**

David Smith

As professional astronomers employ ever-greater instruments with ever-greater costs, is there any way in which amateurs with modest equipment can contribute to the science? Yes, says Peter Dunckel, an amateur astronomer who resides in San Francisco. At his "Rattlesnake Creek Observatory", located at his vacation home in Grass Valley, he gathers asteroid light curves with his 7- and 10-inch telescopes, and combines the results with those from other observatories.

The fruits of this amateur-professional collaboration include Peter's first scientific paper at age 75, after a career as a marketing executive. *Figure of the double Asteroid 90 Antiope from adaptive optics and lightcurve observations* appeared in the April 2007 edition of *Icarus*, with Peter's name among 20 authors, and his Rattlesnake Creek Observatory included with 8 other contributing institutions. Peter is pleased with the results from the combination of his observational data with those from Lick, Keck, and VLT.

Peter will present results from the Antiope mapping project at the October 27 general meeting. He will then use that as a springboard to promote the opportunities for further amateur/professional collaborations, about which he is excited.

A news release on the 90 Antiope project can be found at [http://www.berkeley.edu/news/media/releases/2007/03/29\\_antiope.shtml](http://www.berkeley.edu/news/media/releases/2007/03/29_antiope.shtml).



Peter Dunckel. See more pictures related to this talk on pages 6 and 7. Photo courtesy of Peter Dunckel.

# DEEP SKY OBSERVING

by Mark Wagner

October 2007 third quarter to new moon observing list The list begins in the north and moves southward. Objects are within roughly a two hour section of right ascension that is at a comfortable elevation to the east at astronomical dark. This list is just a sampling of the full list which is at <http://www.resource-intl.com/Deep.Sky.Oct.07.html>.

| Object   | Const.                                                                        | Type | Size        | Mag  | R.A.     | Dec.     |
|----------|-------------------------------------------------------------------------------|------|-------------|------|----------|----------|
| NGC 40   | Cep                                                                           | PN   | 70.0"x60.0" | 10.7 | 00 13 00 | 72 31 19 |
|          | <i>380x provided a nice view with subtle irregularities in the interior</i>   |      |             |      |          |          |
| N7762    | Cep                                                                           | OC   | 11.0'       | 10   | 23 49 54 | 68 01 00 |
|          | <i>About 100 stars in a 20' field including many mag 12-13 stars.</i>         |      |             |      |          |          |
| M52      | Cas                                                                           | OC   | 12.0'       | 6.9  | 23 24 49 | 61 35 51 |
|          | <i>150-175 stars in a 15' diameter; beautiful, appears fully resolved.</i>    |      |             |      |          |          |
| NGC 7510 | Cep                                                                           | OC   | 4.0'        | 7.9  | 60 34 08 | 60 34 08 |
|          | <i>Very small tight knot of 10 bright stars in a fan or comet shape.</i>      |      |             |      |          |          |
| N7419    | Cep                                                                           | OC   | 6.0'        | 13   | 22 54 19 | 60 48 47 |
|          | <i>Small group of 25-30 stars to mag 15, elongated 3'x1' NW-SE</i>            |      |             |      |          |          |
| N7354    | Cep                                                                           | PN   | 36.0"       | 12.9 | 22 40 20 | 61 17 06 |
|          | <i>Direct vision 70x, 260x gave fairly defined perimeter, maybe green.</i>    |      |             |      |          |          |
| NGC 7380 | Cep                                                                           | OC   | 12.0'       | 7.2  | 22 47 00 | 58 06 00 |
|          | <i>Triangular overall shape, 14 or 15 brighter members.</i>                   |      |             |      |          |          |
| NGC 7296 | Lac                                                                           | OC   | 4.0'        | 9.7  | 22 27 57 | 52 18 58 |
|          | <i>Fairly conspicuous, bright stars overlain with a nebulous haze.</i>        |      |             |      |          |          |
| NGC 7789 | Cas                                                                           | OC   | 15.0'       | 6.7  | 23 57 26 | 56 43 14 |
|          | <i>Remarkably rich and fairly uniform carpet of mag 11 and fainter stars.</i> |      |             |      |          |          |
| NGC 7686 | And                                                                           | OC   | 14.0'       | 5.6  | 23 30 12 | 49 07 00 |
|          | <i>Part of Abell Galaxy Clouster AGC2666 - nice chain of galaxies!</i>        |      |             |      |          |          |
| NGC 7662 | And                                                                           | PN   | 37.0"       | 9.2  | 23 25 53 | 42 32 06 |
|          | <i>Blue Snowball - use as much mag as possible to see detail.</i>             |      |             |      |          |          |
| N7640    | And                                                                           | GX   | 11.6'x1.9'  | 11.9 | 23 22 06 | 40 50 43 |
|          | <i>A large almost edge-on spiral, bright overall, low surface brightness.</i> |      |             |      |          |          |
| NGC 7331 | Peg                                                                           | GX   | 14.5'x3.7'  | 9.4  | 22 37 04 | 34 25 00 |
|          | <i>Large detailed spiral galaxy with several dim companions.</i>              |      |             |      |          |          |
| HCG 92   | Peg                                                                           | GX   | 14.5'x3.7'  | 9.4  | 22 37 04 | 34 25 00 |
|          | <i>Arp 319 - Stephan's Quintet. This is the most famous Hickson.</i>          |      |             |      |          |          |
| N7457    | Peg                                                                           | GX   | 4.3'x2.3'   | 12.1 | 23 00 59 | 30 08 41 |
|          | <i>Viewed from suburban backyard in 10" scope.</i>                            |      |             |      |          |          |
| Arp 46   | Peg                                                                           | GX   | 1.4'x0.8'   | 13.5 | 23 33 49 | 30 03 37 |
|          | <i>Two UGC galaxies (12665 &amp; 12667), connected spiral arms.</i>           |      |             |      |          |          |
| Arp 86   | Peg                                                                           | GX   | 3.3'x2.0'   | 12.8 | 23 47 04 | 29 29 02 |
|          | <i>NGC 7752 and 7753 - difficult view in 12.5"</i>                            |      |             |      |          |          |

Note: Source catalogs are Messier, Arp, Abell Planetary, Abell Galaxy Cluster (AGC), Hickson Compact Galaxy (HCG), Herschel 400-I, Herschel 400-II. Herschel 400-I are identified as NGCXXXX, Herschel 400-II as NXXXX.

## Breaking Up is Hard to Do

Akkana Peck

Mars is finally getting close enough to show some detail as it draws nearer to its opposition at the end of the year. By late October its apparent size grows to a bit over 11" (that's arc seconds, not inches). That's still not large, but it's big enough that a determined late-night observer can start picking out major surface features like the poles, Syrtis Major and Hellas. Look for the phase, too: it'll be noticeably gibbous.

Unfortunately, it's a very late evening object throughout the month. Even by the end of October Mars doesn't rise until just before 10pm, so it's not a good bet if you like to set up a telescope on Halloween for the trick- or-treaters. Halloween presents a puzzle this year. Most years, it's dark enough to show anything in the sky by the time the kids come calling, because we've already switched from Daylight Savings Time back to standard time. But this year, thanks to the Energy Policy Act of 2005, we'll stay on Daylight Savings Time through the first weekend in November, so the little witches and warlocks may start coming by while the sky is still light.

The gibbous near-third-quarter moon won't help much either -- it rises even later than Mars. For early trick-or-treaters, Jupiter is your best bet: it sets early (around 8:20) but if you have a good western horizon, even down low Jupiter can be impressive to folks who haven't looked through a telescope before. Once it sets, you could try Uranus: it's well up by sunset, and its pale green disk might impress some trick- or-treaters. Neptune, too, is up and visible in the early evening sky.

Pluto, obviously not a good trick-or-treater showpiece, isn't a great target for anyone this month, since it starts the evening low in the sky and sets early. The rest of the planets are up

in the morning: Venus, Saturn, and low, fleeting Mercury.

If you get up early to look at those planets, mid to late October is a good time to watch for the morning zodiacal light. Look to the east just before the sky begins to brighten. The faint glow comes from tiny particles following the plane of the ecliptic, left over from the dust of the early solar system. It's a band of light reaching up from the horizon along the ecliptic, past Venus, Saturn and the star Regulus.

Speaking of the early solar system, did you hear the news last month about the Baptistina Breakup? No, not a band: a big asteroid. A recent study of asteroid (298) Baptistina suggests that it's

– creating a swarm of asteroids now called the Baptistina family.

The team publishing in *Nature* ran computer simulations on the orbits of the currently known Baptistinas.

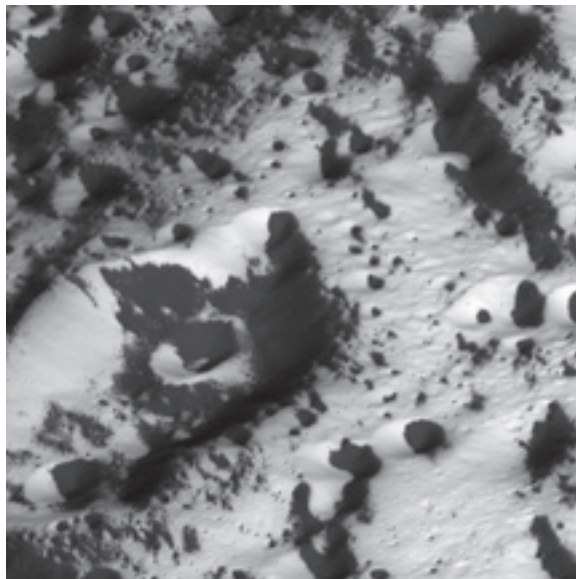
They think the breakup occurred roughly 160 million years ago, in the mid-Jurassic. Most of the fragments stayed in

the asteroid belt, but a few of them wandered gradually in toward the inner planets; and almost 100 million years later, one of the fragments from the breakup smacked down near the modern-day Yucatan peninsula as the Chicxulub meteor, the impact that most likely wiped out the dinosaurs.

But besides being interesting, does this have anything to do with shallow sky observing? Right now, (298) Baptistina itself isn't much to look at. It's 15th magnitude and sitting in the belly of Leo, rising at about 4am.

But it turns out there's another member of the Baptistina family whose effects you've seen many times. The research team thinks that some 45 million years before that Chicxulub impact, another piece from the same parent asteroid smacked into the moon – to become the crater Tycho, with its huge ray system that's so prominent when the moon is full.

So if you find yourself looking at the full moon around the end of this month, or if any little dinosaurs come to your door on the 31st demanding candy, give a thought to little (298) Baptistina ... sister to the asteroid that killed the dinosaurs and gave us mammals a chance.



*This picture of the surface of Iapetus was taken September 10, 2007 as Cassini made its closest approach to this Saturnian moon. Iapetus is very light in one hemisphere and very dark in the other. This image shows an area near where the two hemispheres meet. Photo courtesy of NASA/JPL/STSci.*

actually just a piece of a much larger 170 km asteroid. This parent asteroid broke up long ago – probably due to a collision with another asteroid



Cockroaches are supposed to be tough, able to survive anything from a good stomping to a nuclear blast. But roaches are wimps compared to a little molecule that has recently caught the eye of biologists and astronomers—the polycyclic aromatic hydrocarbon.

Polycyclic aromatic hydrocarbons (PAHs for short) are ring-shaped molecules made of carbon and hydrogen. “They’re all around us,” says Achim Tappe of the Harvard Center for Astrophysics. “PAHs are present in mineral oils, coal, tar, tobacco smoke and automobile exhaust.” Aromatic, ring-shaped molecules structurally akin to PAHs are found in DNA itself!

That’s why Tappe’s recent discovery may be so important. “PAHs are so tough, they can survive a supernova.”

The story begins a few thousand years ago when a massive star in the Large Magellanic Cloud exploded, blasting

nearby star systems and interstellar clouds with hot gas and deadly radiation. The expanding shell, still visible from Earth after all these years and catalogued by astronomers as “N132D,” spans 80 light years and has swept up some 600 Suns worth of mass.

Last year “we observed N132D using NASA’s Spitzer Space Telescope,” says Tappe. Spitzer is an infrared (IR) telescope, and it has a spectrometer onboard sensitive to the IR emissions of PAHs. One look at N132D revealed “PAHs all around the supernova’s expanding shell. They appear to be swept up by a shock wave of 8 million degree gas. This is causing some damage to the molecules, but many of the PAHs are surviving.”

Astronomers have long known that PAHs are abundant not only on Earth but throughout the cosmos—they’ve been found in comet dust, meteorites and many cold interstellar clouds—but

who knew they were so tough? “This is our first evidence that PAHs can withstand a supernova blast,” he says.

Their ability to survive may be key to life on Earth. Many astronomers are convinced that a supernova exploded in our corner of the galaxy 4-to-5 billion years ago just as the solar system was coalescing from primitive interstellar gas. In one scenario of life’s origins, PAHs survived and made their way to our planet. It turns out that stacks of PAHs can form in water—think, primordial seas—and provide a scaffold for nucleic acids with architectural properties akin to RNA and DNA. PAHs may be just tough enough for genesis.

Cockroaches, eat your hearts out.

Find out about other Spitzer discoveries at [www.spitzer.caltech.edu](http://www.spitzer.caltech.edu).

*This article was provided by the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration.*



*Using the IR spectrometer on the Spitzer Space Telescope, scientists found organic molecules in supernova remnant N132D. Photo courtesy of NASA.*

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# CALSTAR

Rob Hawley

It's almost time for the annual CalStar Star Party! This year it will be held the nights of October 11 - 13 at Lake San Antonio. Reservations are now open. LSA is one of the darkest areas in California. Yet it is just a 3 hour drive from San Jose.

We structure this event to support observers at all levels of experience. If you are comfortable with draconian light restrictions then we have a dark enforced area that will reward you by preserving your night vision to get that elusive glob.

If you are new to star parties (or want a more casual experience) then you should consider the casual area. Someone that has not been to an event

like this before should read <http://www.sjaa.net/calstar/info.html>

Some visitors prefer to only visit for a portion of the night and then leave. This year we have a new Late Arrival/Early Departure area which uses a part of the park we have not previously used. It is a different entrance that is easy to find at night, easy to drive into and out of after darkness, has good horizons, and is only a short walk from the food and other observers. Using this will reduce the impact of arrivals and departures on the more controlled areas.

We will sell BBQ meals (by prepayment) for Friday and Saturday nights. See the website for more details.

This year all visitors will receive a placard (really a piece of paper) with their name on it. For most this will be a part of the registration process. No worries if you are not sure which vehicle you will be using. There will be a registration canopy set up at LSA where you can get (or replace) your placard.

Remember the dark and casual areas are closed 8 PM to twilight to all vehicle traffic. Anyone arriving after that time (or wanting to leave before morning) need to use the LA / ED area. Signs will direct you.

To register, for more information, or to get a tour of the new layout visit our website at [www.sjaa.net/calstar](http://www.sjaa.net/calstar).

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## The Last 31 Days In Astronomy

The news seen between August 10 and September 10, 2007.

AUG-20-2007 **Nearest Neutron Star** Astronomers from Penn State University report that they may have found the closest neutron star. The distance is between 250 and 1,000 lightyears. The object, nicknamed Calvera, will be the subject of a lot of study including some observations from Chandra to see if it pulsates in X-rays. <http://www.science.psu.edu/alert/Fox8-2007.htm>

AUG-29-2007 **Deep, Dark Pit on Mars** The Mars Reconnaissance Orbiter has returned some high-resolution images. One of them confirms the appearance of a crater that appears to be deep and nearly perfectly vertical. These so-called "pit craters" have been seen on volcanoes in Hawaii. <http://space.newscientist.com/article.ns?id=dn12566>

SEP-04-2007 **Phoenix Passes Tests** The Phoenix Mars lander has passed earlier tests of its camera, radar and UHF transmitter. <http://www.jpl.nasa.gov/news/news.cfm?release=2007-094a>

SEP-06-2007 **Days (on Saturn) really are shorter** Calculations based on radio astronomy performed on Cassini, Voyager and Pioneer spacecraft have been used to come up with a new estimate for the length of a Saturnian day. The new estimate is shorter than the previous estimate by about 15 minutes. The new estimate is 10 hours, 32 minutes and 35 seconds. What difference does that make? Well, the longer day implied that all of the winds on the planet must all be traveling in the same direction. The shorter day allows for more complex weather. [http://www.space.com/scienceastronomy/070906\\_saturn\\_day.html](http://www.space.com/scienceastronomy/070906_saturn_day.html)

SEP-06-2007 **Rocket Failure Compacts Schedule** A Proton rocket launching a Japanese satellite failed dropping rocket parts and fuel over the Kazakh countryside. Besides creating more tension between Russia and Kazakhstan, the satellite launching schedule is getting cramped. The Atlas 5 is booked until 2009 at the earliest. Sea Launch is booked through 2008, assuming it can resume launches this year. Ariannospace might have a slot open in 2008. [http://www.space.com/spacenews/070910\\_busmon\\_protonfail.html](http://www.space.com/spacenews/070910_busmon_protonfail.html)

SEP-07-2007 **Rovers Revived** The Mars rovers are ready to go back to work after surviving dust storms. Opportunity will travel into an 800 meter wide crater called Victoria Crater but early results suggest that the terrain might be too slippery. <http://www.jpl.nasa.gov/news/news.cfm?release=2007-098>, [http://www.space.com/news/070912\\_rovers\\_update.html](http://www.space.com/news/070912_rovers_update.html), <http://marsrovers.jpl.nasa.gov/newsroom/pressreleases/20070907a.html>

## Asteroids that Hit the Earth - October 3, 2007

Andrew Fraknoi

On Wednesday, Oct. 3rd, 2007, at 7 pm, Astronomer David Morrison of NASA's Ames Research Center will give a non-technical, illustrated talk on: "Taking a Hit: Asteroid Impacts and Evolution" as part of the Silicon Valley Astronomy Lectures in the Smithwick Theater, Foothill College, El Monte Road and Freeway 280, in Los Altos Hills, California.

Free and open to the public. Parking on campus costs \$2.

Call the series hot-line at 650-949-7888 for more information and driving directions.

No background in science will be required for this talk.

Asteroids have hit the Earth many times in the past, and they will continue to hit in the future, whether we are prepared or not. Collisions with our planet over 4.5 billion years have profoundly influenced the evolution of life. In fact, were it not for the impact of a 15-km

wide asteroid 65 million years ago, it is likely humanity would not be here.

Impacts are important for our future as well as our past. In the last two decades we have learned not only how to evaluate the impact hazard but also (in principle) how to defend ourselves. The astronomers operating the Spaceguard Survey of Near Earth Asteroids have already reduced the risk of fatality from unknown asteroids by at least 75%. Unlike other natural hazards, we now have the capability of removing most of the impact risk within the next generation. However, the government still does not have a plan of action for when an asteroid is discovered heading our way or when an impact happens without any warning.

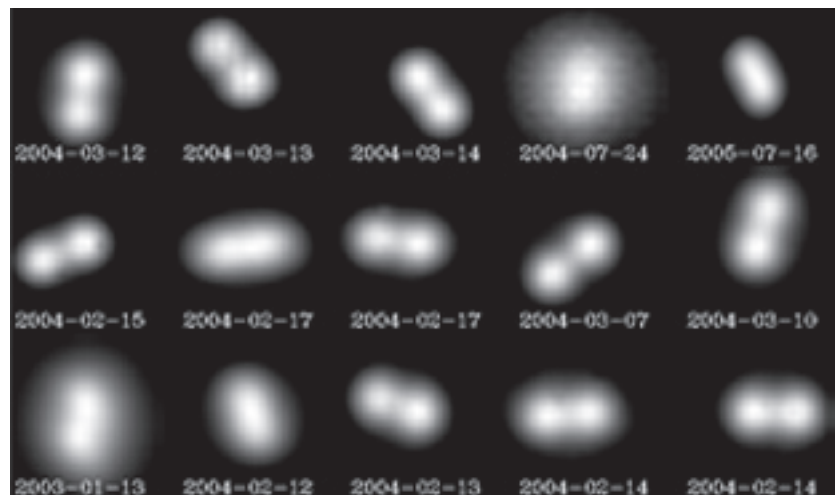
David Morrison is one of the world's experts on the study of asteroid impacts. He is the Senior Scientist at the NASA Astrobiology Institute, where he participates in a variety of research programs in the study of the living universe. Dr. Morrison is the author of

more than 155 technical papers and has published a dozen books (including several widely used college textbooks in astronomy). He is the recipient of numerous awards for his scientific and his educational work, including the Sagan Medal of the American Astronomical Society for public communication. Dr. Morrison was a founder of the multi-disciplinary field of astrobiology. Asteroid 2410 Morrison is named in his honor, but he assures us that it is not one of those that might hit the Earth.

The lecture is co-sponsored by:

- \* NASA Ames Research Center
- \* The Foothill College Astronomy Program
- \* The SETI Institute
- \* The Astronomical Society of the Pacific.

Past Silicon Valley Astronomy Lectures are now available in MP3 format at: <http://www.astrosociety.org/education/podcast/index.html>



The Double Asteroid (90) Antiope  
(NACO/VLT)

ESO Press Photo 18a/07 (29 March 2007)

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Peter Duncel's Rattlesnake Creek Observatory is shown on the left. Photo Courtesy of Peter Duncel. Above is the double asteroid 90 Antiope viewed in various orientations. Photo courtesy of ESO. A better view is available at [http://www.planetary.org/explore/topics/near\\_earth\\_objects/asteroids\\_and\\_comets/antiope.html](http://www.planetary.org/explore/topics/near_earth_objects/asteroids_and_comets/antiope.html).



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### *Telescope Loaner Program*

(see elsewhere this page)

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## *From the President of the club*

## Changes to the Telescope Loaner Program

I am happy to announce that the board is turning on the loaner program. The board asked me to be the acting chair. We need lots of help as will be described later in this message.

To avoid chaos we will turn on the program in stages. I hope that it will not be long before the full program will be active, but I cannot tell you how long that will be.

A number of the scopes are currently being stored by non-members. Initially only these scopes are available immediately. You can see which scopes are available at <http://www.sjaa.net/loaners/sjaaloan.html>

This will allow the club to get the scopes back under control. Storage for these scopes is a major issue. Even if you only want to donate space for storage, we would appreciate you taking the loan of one of these scopes.

Once we get these scopes under control, we will open the loans up to other scopes.

The new loaner program will depend on the efforts of volunteers. We need help in the following areas:

1. Storage
2. Avoiding the high cost of commercial space
3. Some Place to hold returned scopes
4. Handling the transfer of scopes from storage to loanee and vice versa
5. Assistance in evaluating and documenting current inventory
6. Training new users

To borrow (provide storage for) one of the available scopes or for more information please contact me at [loaner@sjaa.net](mailto:loaner@sjaa.net).

*Rob Hawley*

### **Publication Statement**

*SJAA Ephemeris*, newsletter of the San Jose Astronomical Association, is published monthly.

San Jose Astronomical Association,  
P.O. Box 28243  
San Jose, CA 95159-8243

### **Submit**

Submit articles for publication in the *SJAA Ephemeris*. Send articles to the editors via e-mail to [ephemeris@sjaa.net](mailto:ephemeris@sjaa.net). **Deadline, 10th of previous month.**



The Antiope Doublet  
(Artist's Impression)

ESO Press Photo 18a/07 (29 March 2007)

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To the left is an artist's conception of the double asteroid 90 Antiope. Photo courtesy of ESO. ESO photos are copyrighted.



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## San Jose Astronomical Association Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

☐ **New**    ☐ **Renewal** (Name only if no corrections)

☐ **I'll get the Ephemeris newsletter online**

<http://ephemeris.sjaa.net> Questions?

Send e-mail to [membership@sjaa.net](mailto:membership@sjaa.net)

### Membership Type:

- ☐ Regular — \$20  
☐ Regular with Sky & Telescope — \$53  
☐ Junior (under 18) — \$10  
☐ Junior with Sky & Telescope — \$43

Bring this form to any SJAA Meeting  
or send to the club address (above).

Please make checks payable to "SJAA".

Subscribing to Sky & Telescope magazine through the SJAA  
saves you \$10 off the regular rate. (S&T will not accept multi-year  
subscriptions through the club program. Allow 2 months lead time.)

You can join or renew online:

<http://www.sjaa.net/SJAAmembership.html>

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\_\_\_\_\_

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\_\_\_\_\_

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\_\_\_\_\_

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\_\_\_\_\_